Nichtparametrische Methoden für die Sozialwissenschaften

JEFF GILL
Department of Political Science
Division of Biostatistics
Department of Surgery (Public Health Sciences)
Washington University, St. Louis

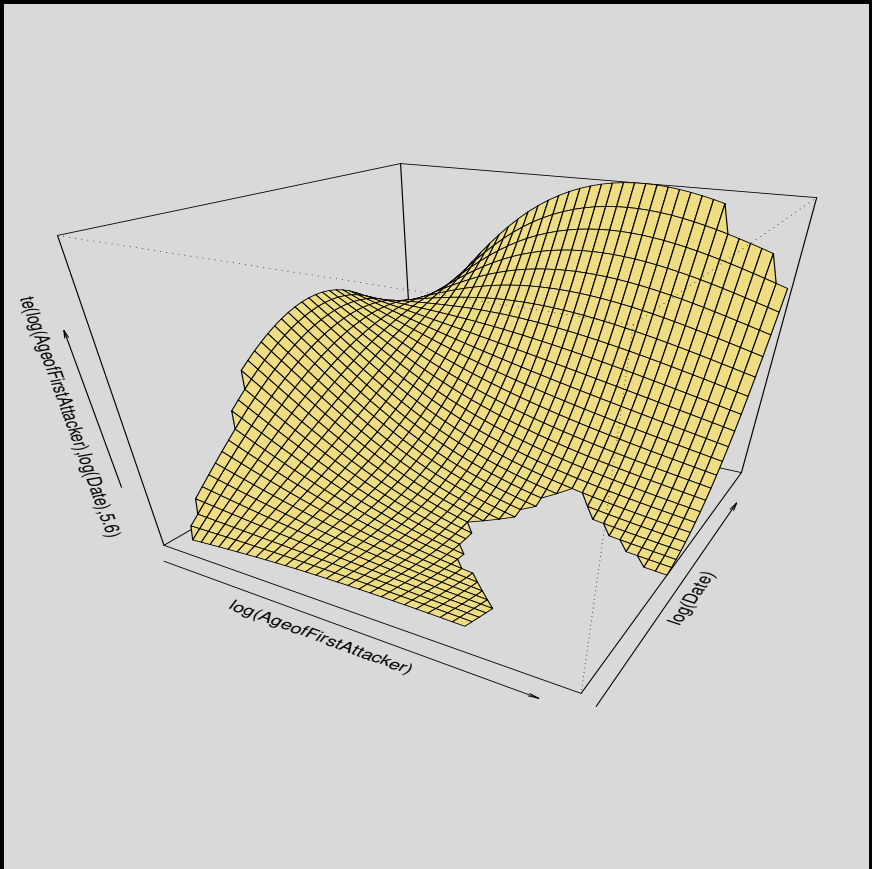
Graphing the Terrorism Data Analysis

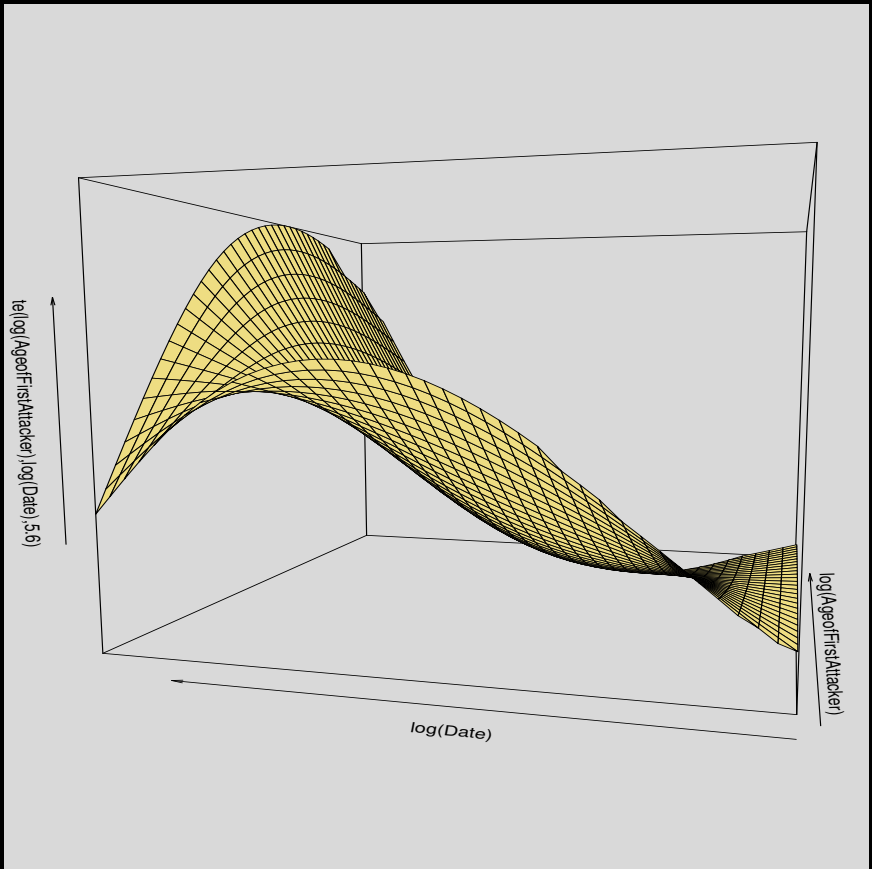
- There is a handy but slightly confusing plot routine for GAMs:

```
postscript("Class.Stat.Comp/harr.persp1.ps")
par(mfrow=c(1,1),mar=c(1,1,0,1),oma=c(0,0,0,0),col.axis="white",col.lab="white",
     col.sub="white",col="black",bg="grey85",cex.lab=3)
plot.gam(harr.gam3,too.far=0.25,lwd=1,pers=TRUE,col="lightgoldenrod",cex.lab=1.3)
dev.off()
postscript("Class.Stat.Comp/harr.persp2.ps")
par(mfrow=c(1,1),mar=c(1,2,0,1),oma=c(0,0,0,0),col.axis="white",col.lab="white",
     col.sub="white",col="black",bg="grey85",cex.lab=3)
plot.gam(harr.gam3,too.far=0.25,lwd=1,pers=TRUE,col="lightgoldenrod",cex.lab=1.3,
         theta=285,phi=5)
dev.off()
```

- This option gives the perspective plot.

Graphing the Terrorism Data Analysis



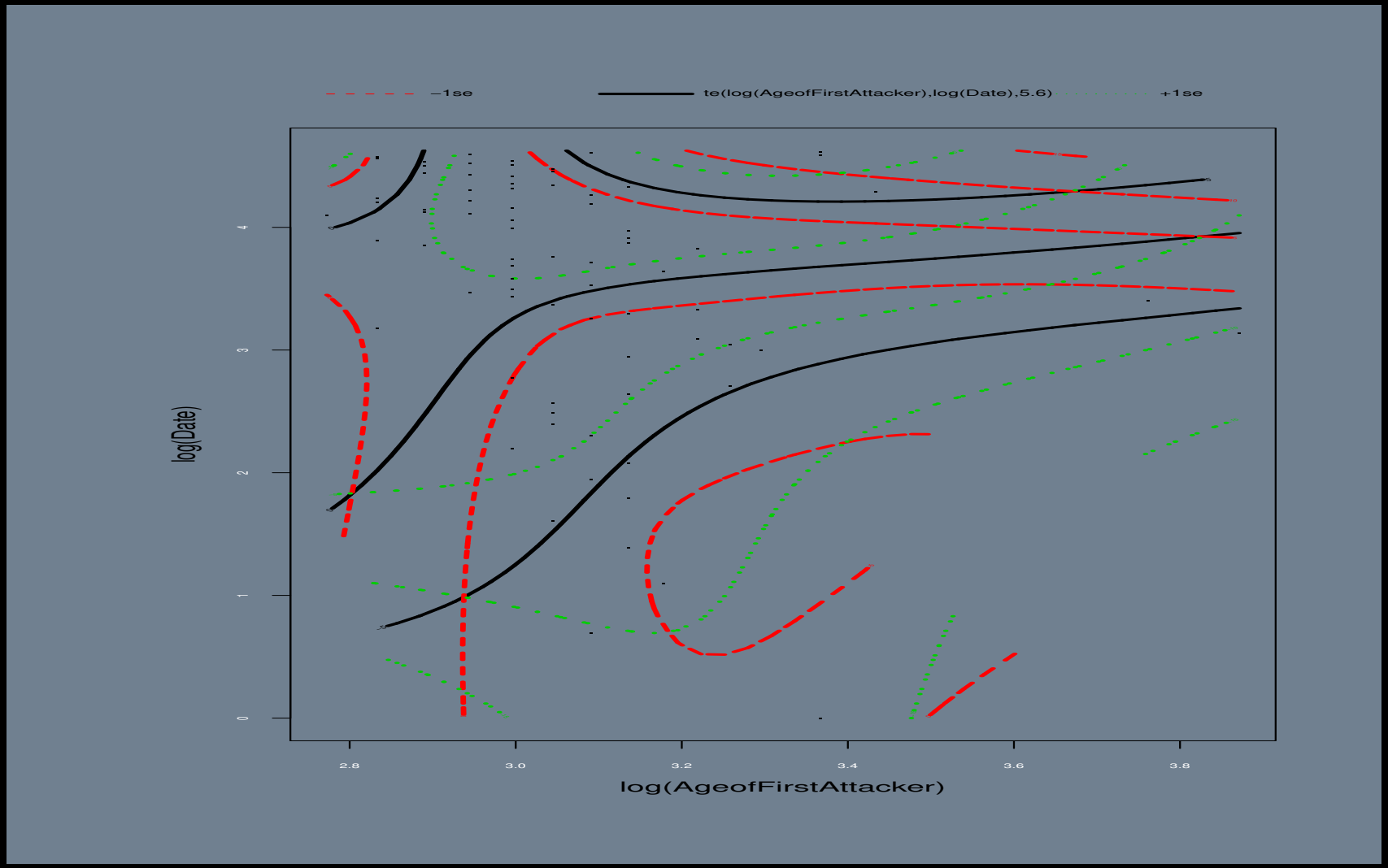


Graphing the Terrorism Data Analysis

- We can also plot the smooth function that results from the bivariate fit:

```
postscript("Class.Stat.Comp/harr.smooth.ps")
par(mfrow=c(1,1),mar=c(8,8,8,3),col.axis="white",col.lab="white",
    col.sub="white",col="black",bg="slategrey")
plot.gam(harr.gam3,too.far=0.25,lwd=3,cex.lab=2,cex.main=1.3)
dev.off()
```

Graphing the Terrorism Data Analysis

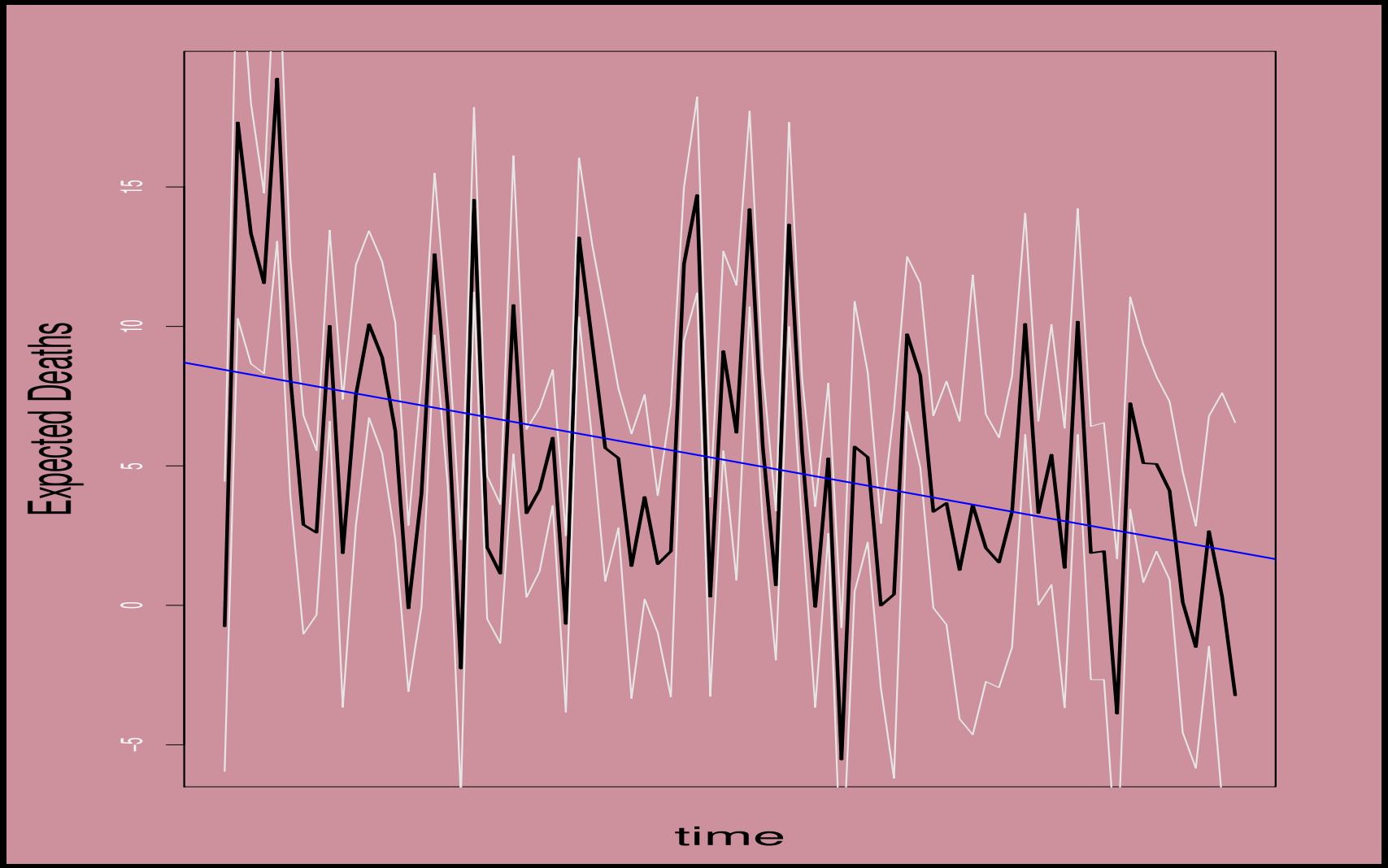


Graphing the Terrorism Data Analysis

- Recall that it is useful to see predictions on the outcome variable.
- Since our LHS is ordered by time, this provides a *serial prediction*.
- Simple process:

```
harr.gam3.predict <- predict(harr.gam3,se=TRUE,type="response")
postscript("Class.Stat.Comp/harr.predict.ps")
par(mfrow=c(1,1),mar=c(5,5,3,3),col.axis="white",col.lab="black",
    col.sub="white",col="black",bg="pink3",cex.lab=2)
plot(harr.gam3.predict$fit,lwd=2,type="l",ylab="Expected Deaths",
     xaxt="n",xlab="time")
lines(harr.gam3.predict$fit+1.96*harr.gam3.predict$se.fit,col="grey90")
lines(harr.gam3.predict$fit-1.96*harr.gam3.predict$se.fit,col="grey90")
y <- as.vector(harr.gam3.predict$fit); x <- 1:78
abline(lm(y ~ x),col="blue",lwd=2)
dev.off()
```

Graphing the Terrorism Data Analysis



GDP Growth

- This example is about how to tell if a GAM fit is better than an LM fit.
- Data used in Sachs and Warner (1997a, 1997b, 1995a, 1995b), eg. "Fundamental Source of Long-Run Growth", American Economic Review (1997b).
- See `http://www.bris.ac.uk/Depts/Economics/Growth/sachs.htm` for more details.
- Variables:
  - `GR6590` Average annual growth in real GDP per economically active population between 1970 and 1989.
  - `OPEN6590` The fraction of years during the period 1965-1990 in which the country is rated as an open economy.
  - `TROPICS` Takes the value 1 for a country in which the entire land area is subject to a tropical climate.
  - `LIFE` Log of life expectancy at birth, circa 1965-1970.
  - `ICRGE80` An average of 5 sub-indexes, each based on survey data from Political Risk Services.
  - `CGB7090` Central government savings is measured as current revenues minus current expenditures of the central government, expressed as a fraction of GDP.

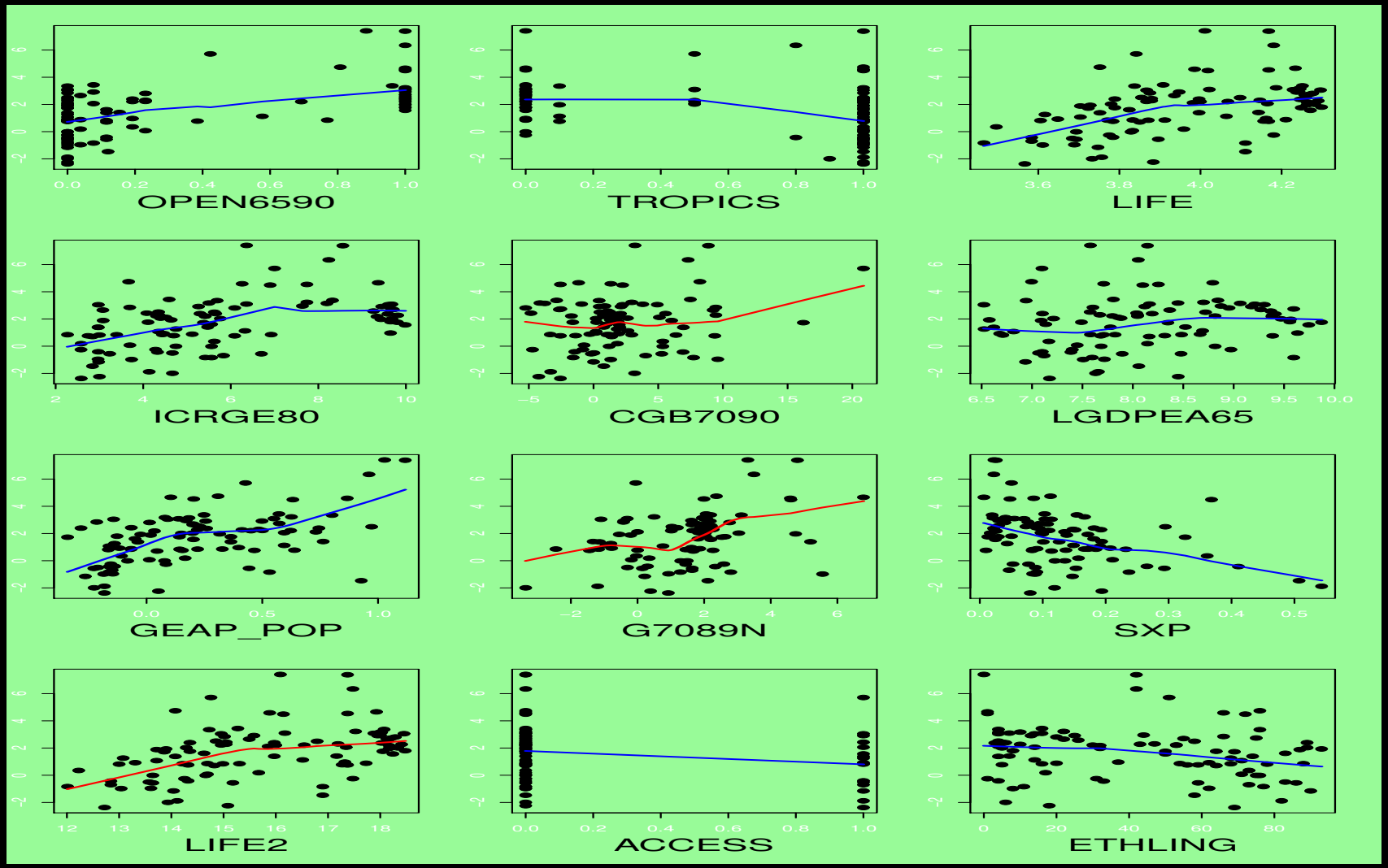
- ▷ `LGDPEA65` Natural log of real (purchasing power parity adjusted) GDP per economically active population.
- ▷ `GEAP_POP` Difference between the growth rate of the economically active population (between ages 15 and 65) and growth of total population.
- ▷ `G7089N` Growth of neighboring countries.
- ▷ `INFL6590` Average inflation 1965-90.
- ▷ `SXP` Share of exports of primary products in GNP in 1970.
- ▷ `LIFE2` Life squared.
- ▷ `ACCESS` Physical access to international waters.
- ▷ `ETHLING` Ethno-linguistic fractionalization taken from related work by Mauro (1995) and Easterly and Levine (1996).

GDP Growth

```
sachs <- read.table("http://jgill.wustl.edu/data/sachs.csv",sep=",",header=TRUE)
library(mice)
imp.sachs <- mice(sachs[,c(3:15)],m=10)
comp.sachs <- complete(imp.sachs,1)
dim(comp.sachs)
95 13

postscript("Class.Stat.Comp/sachs1.ps")
par(mfrow=c(4,3),mar=c(5,2,2,2),col.axis="white",col.lab="black",
    col.sub="white",col="black",bg="palegreen",cex.lab=2)
for (i in c(1:9,11:13))  {
    plot(comp.sachs[,i],comp.sachs[,10],cex=1.25,pch=19,
        xlab=names(comp.sachs)[i],ylab="")
    lo.object <- lowess(comp.sachs[,10]~comp.sachs[,i],f=2/3)
    if (i==5 | i==8 | i==11) lines(lo.object$x,lo.object$y,lwd=2,col="red")
    else lines(lo.object$x,lo.object$y,lwd=2,col="blue")
}
dev.off()
```

GDP Growth, Graphed Against Explanatory Variables



GDP Growth

```
gdp.lm   <- lm(GR6590 ~ OPEN6590 + TROPICS + LIFE + ICRGE80 +
                CGB7090 + LGDPEA65 + GEAP_POP + G7089N +
                SXP + LIFE2 + ACCESS + ETHLING, data=comp.sachs)

gdp.gam1 <- gam(GR6590 ~ OPEN6590 + TROPICS + ICRGE80 +
                s(CGB7090,bs="cr",k=20) + LGDPEA65 + GEAP_POP +
                s(G7089N,bs="cr",k=20) + SXP + s(LIFE2,bs="cr",k=20) +
                ACCESS + ETHLING, data=comp.sachs)

summary(gdp.lm)

summary(gdp.gam1)
```

GDP Growth, LM Results

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.20e+02   3.76e+01   -3.19  0.00202
OPEN6590     1.86e+00   3.33e-01    5.58  3.0e-07
TROPICS     -7.62e-01   2.70e-01   -2.83  0.00589
LIFE         6.54e+01   1.91e+01    3.42  0.00099
ICRGE80      2.75e-01   7.66e-02    3.59  0.00057
CGB7090      1.22e-01   2.07e-02    5.91  7.7e-08
LGDPEA65    -1.88e+00   1.98e-01   -9.50  7.4e-15
GEAP_POP     9.95e-01   3.30e-01    3.02  0.00340
G7089N       9.97e-02   5.85e-02    1.70  0.09226
SXP         -3.00e+00   9.40e-01   -3.19  0.00203
LIFE2       -7.87e+00   2.45e+00   -3.21  0.00192
ACCESS      -6.03e-01   2.47e-01   -2.44  0.01690
ETHLING     -1.86e-03   3.39e-03   -0.55  0.58382

Residual standard error: 0.797 on 82 degrees of freedom
Multiple R-squared: 0.851,        Adjusted R-squared: 0.829
F-statistic: 38.9 on 12 and 82 DF,  p-value: <2e-16
```

GDP Growth, GAM Results

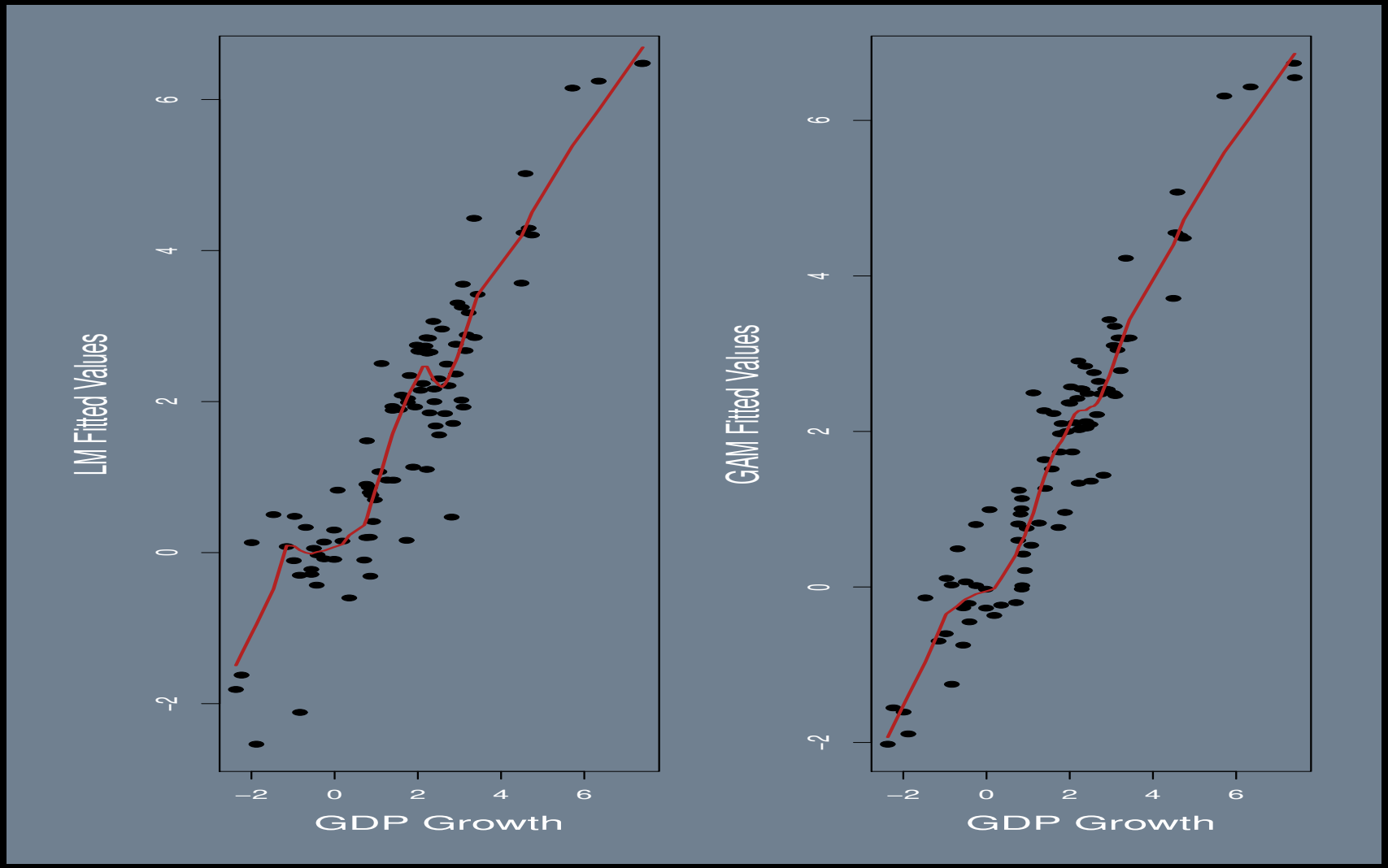
```
Parametric coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 16.83566    1.58590   10.62  5.0e-16
OPEN6590     2.06025    0.31030    6.64  6.5e-09
TROPICS     -0.53903    0.27060   -1.99   0.0504
ICRGE80      0.37233    0.07294    5.10  2.9e-06
LGDPEA65    -2.12289    0.20046  -10.59  5.6e-16
GEAP_POP     0.64165    0.37855    1.70   0.0947
SXP         -2.74615    0.87199   -3.15   0.0024
ACCESS      -0.69192    0.22716   -3.05   0.0033
ETHLING     -0.00329    0.00314   -1.05   0.2977

Approximate significance of smooth terms:
             edf Ref.df     F p-value
s(CGB7090)  3.26   3.26 16.64 1.5e-08
s(G7089N)  11.47  11.47  1.92    0.05
s(LIFE2)    3.89   3.89 12.11 2.3e-07

R-sq.(adj) =  0.881   Deviance explained = 91.5%
GCV score = 0.62295  Scale est. = 0.44187   n = 95
```

How Do We Know Which Is Better?

- Wald tests, worse for GAM: TROPICS, GEAP_POP; worse for LM: ETHLING; the rest have very small p-values for both.
- Scale, GAM: $\sigma^2 = 0.442$, LM: $\sigma^2 = 0.797$.
- Variance explained, GAM: $R^2 = 0.881$, LM: $R^2 = 0.851$.
- Graphing predictors:

```
postscript("Class.Stat.Comp/sachs2.ps")
par(mfrow=c(1,2),mar=c(5,5,1,1),oma=c(1,1,1,1),col.axis="white",col.lab="white",
    col.sub="white",col="black",bg="slategrey",cex.lab=1.5)
plot(comp.sachs$GR6590,gdp.lm$fitted.values,cex=1.05,pch=19,
    xlab="GDP Growth",ylab="LM Fitted Values")
lo.object <- lowess(gdp.lm$fitted.values ~ comp.sachs$GR6590,f=1/5)
lines(lo.object$x,lo.object$y,lwd=2,col="firebrick")
plot(comp.sachs$GR6590,gdp.gam1$fitted.values,cex=1.05,pch=19,
    xlab="GDP Growth",ylab="GAM Fitted Values")
lo.object <- lowess(gdp.gam1$fitted.values ~ comp.sachs$GR6590,f=1/5)
lines(lo.object$x,lo.object$y,lwd=2,col="firebrick")
dev.off()
```

GDP Growth, Fit Comparison